



SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED

OFFICIAL DEFENCE CAPABILITY PRESENTATION • 5-PAGE EXECUTIVE PUBLICATION

INDIA-MADE DEFENCE LOGISTICS VTOL & TACTICAL OVERWATCH UAS

MODULAR HYBRID SYSTEM

ZERO-RUNWAY VTOL

HIGH-ALTITUDE LAC / LOC RESUPPLY

BVLOS SWARM CAPABLE

AEROSPACE DEFENCE PLATFORM

SHIV-NETRA47 VTOL HYBRID

Twin-Boom Aerodynamic Carbon Fuselage • Quad Vertical Lift • High-Speed Pusher Cruise



180 MIN

FLIGHT ENDURANCE

150 KM

RADIO NETWORK RANGE

5,500 M

SERVICE CEILING (AMSL)

200 KG

SCALABLE CLASS TARGET

BVLOS AUTONOMOUS CRUISE

NAVIC / GPS DUAL-GUIDANCE • TERRAIN FOLLOWING

STRATEGIC OPERATIONAL DOCTRINE & MANDATE

LAC / LOC BORDER FIT

An indigenous, runway-independent unmanned aerial logistics platform built on European flight-proven VTOL engineering, specifically optimized for **harsh Himalayan border resupply (LAC/LoC)**. Combines zero-infrastructure vertical takeoff with high-efficiency fixed-wing cruise for rapid autonomous delivery of critical medical packs, munitions, and persistent tactical ISR overwatch.

- ✓ ZERO-INFRASTRUCTURE VTOL AIRLIFT
- ✓ 5,500M SERVICE CEILING (AMSL)
- ✓ SECURE MANET MESH & BVLOS CRUISE

SOVEREIGN DEFENCE PORTAL

VERIFIED REGISTRY



SCAN TO ACCESS EXECUTIVE PORTAL

[HTTPS://SHIVNETRA47.TECH](https://shivnetra47.tech)

AUTHENTICATED IDEX DOSSIER • EASA / DGCA READY

PILLAR 01 • MISSION CONCEPT

PAGE 02

OPERATIONAL DOCTRINE & RESUPPLY

Runway-free tactical airlift for isolated high-altitude forward posts. Automated 5-stage logistics workflow: Load, Launch, Cruise, Deliver, and Return with quick-change container bays.

ZERO RUNWAY | 5-STAGE WORKFLOW | LAC / LOC FIT

PILLAR 02 • ARCHITECTURE

PAGE 03

MODULAR LRUS & DUAL PLATFORMS

6 swappable line-replaceable modules: composite airframe, flight controller, mission computer, secure comms, modular cargo pod, and rugged GCS. Supporting both VTOL Hybrid and Heavy-Lift Multirotor.

6 SWAPPABLE LRUS | VTOL • MULTIROTOR | SAIL III HERITAGE

PILLAR 03 • MAKE IN INDIA

PAGE 04

LOCALIZATION, TRAINING & MRO

Structured 12-month technology transfer with BCN Drone Center (Spain). Turnkey drone construction and pilot training programmes, domestic manufacturing, and sovereign lifecycle sustaiment.

BCN ECOSYSTEM | PILOT TRAINING | SOVEREIGN MRO

PILLAR 01 // OPERATIONAL ADVANTAGE

HIGH-ALTITUDE FORWARD RESUPPLY & TACTICAL LOGISTICS

Runway-Independent Autonomous Heavy-Lift Airlift for Forward Himalayan Posts (LAC / LoC)


Ecosystem Partner: **BCN Drone Center (Spain)**

TACTICAL FIELD INTEGRATION TRIAL

FIGURE 1.0: FORWARD OPERATING BASE (FOB) RESUPPLY TRIAL — 4,800M AMSL



RAPID MODULAR CARGO POD INTEGRATION AT HIGH ALTITUDE
Field technicians and forward personnel conducting rapid pod exchange in sub-zero alpine conditions. The hermetically sealed bay accommodates emergency blood bags, tactical communications batteries, and precision resupply payloads.

<90 SEC
POD SWAP TIME

5-STAGE AUTONOMOUS LOGISTICS WORKFLOW

END-TO-END AUTOMATED MISSION CYCLE

STAGE 01
PRE-FLIGHT LOAD
Quick-lock mounting of insulated medical containers, ammunition canisters, or dual tactical batteries without specialized tooling.

STAGE 02
ZERO-RUNWAY VTOL
Four high-torque carbon rotors execute vertical liftoff from unprepared mountain clearings, transitioning to fixed-wing flight at 50 kts.

STAGE 03
ALPINE CRUISE
High-efficiency rear pusher motor cruises along pre-programmed mountain valley corridors at 110 km/h with active terrain avoidance.

STAGE 04
AUTONOMOUS DELIVERY
Precision hover descent or autonomous pod release at isolated forward posts with sub-meter GPS/NAVIC coordinates.

STAGE 05
SECURE BVLOS RTB
Automated return-to-base along alternate terrain-masked tracks with low thermal and acoustic signature throughout.

SPEC 01 • ENVIRONMENT

-20°C TO +50°C

HIGH-ALTITUDE CEILING
Engineered for density altitudes exceeding 5,500m AMSL. Carbon-structure heating prevents icing on leading edges and pitot tubes during severe Himalayan winter blizzard operations.

5,500M CEILING | ANTI-ICING | HIMALAYAN FIT

SPEC 02 • MOBILITY

ZERO FOOTPRINT

RUNWAY-FREE LAUNCH
Requires no catapults, runways, or recovery nets. Can deploy directly from forward outpost helipads, ridge trenches, or tactical 4x4 flatbed mobile support units.

ZERO RUNWAY | RIDGE DEPLOYMENT | 4X4 COMPATIBLE

SPEC 03 • PAYLOAD

MODULAR POD

HERMETIC CARGO BAY
Aerodynamic composite fuselage accommodates swappable cargo modules: active temperature-regulated medical pods, tactical supplies, or EO/IR surveillance gimbals.

RAPID SWAP | COLD-CHAIN FIT | TACTICAL PAYLOAD

SPEC 04 • AVIONICS

MANET MESH

SECURE BVLOS LINK
Frequency-hopping spread spectrum (FHSS) telemetry with AES-256 military-grade encryption. Seamless failover between NAVIC satellite navigation and optical inertial dead-reckoning.

AES-256 | NAVIC DUAL-BAND | ANTI-JAM MESH

PILLAR 02 // DUAL-AIRFRAME DEFENCE ECOSYSTEM

MODULAR AIRFRAME ARCHITECTURE & SPECIFICATIONS

Dual-Platform Capability: High-Endurance VTOL Hybrid and Precision Heavy-Lift Multirotor



PLATFORM 01 • SHIV-NETRA47 VTOL HYBRID

ENDURANCE LOGISTICS & ISR

180 MIN
ENDURANCE

150 KM
RADIO RANGE

5,500 M
CEILING

200 KG
MTOW TARGET

Twin-boom aerodynamic carbon airframe combining zero-runway quad-rotor vertical takeoff with high-speed fixed-wing pusher cruise. Optimized for high-altitude mountain corridor resupply and over-the-horizon border surveillance.

PLATFORM 02 • SHIV-NETRA47 TACTICAL MULTIROTOR

PRECISION TACTICAL CARGO

45 MIN
HOVER TIME

25 KG
USEFUL PAYLOAD

4,500 M
CEILING

6 ROTOR
REDUNDANCY

Heavy-lift hexacopter with folding carbon arms designed for pinpoint vertical delivery into restricted mountain trenches, artillery munitions airlift, winch deployment, and low-altitude high-resolution 3D LIDAR terrain scanning.

6 SWAPPABLE LINE-REPLACEABLE UNIT (LRU) SUBSYSTEMS

MODULAR DEFENCE ARCHITECTURE

LRU 01
AIRFRAME CORE

Full pre-preg carbon fiber monocoque fuselage with IP67 all-weather sealing.

LRU 02
TRIPLE AUTOPILOT

Triple-redundant IMU avionics with dual NAVIC/GPS and optical flow fallover.

LRU 03
PROPULSION

Brushless vertical lift motors + heavy pusher cruise motor with FOC telemetry.

LRU 04
TACTICAL COMMS

AES-256 encrypted FHSS datalink with MANET swarm mesh relay capability.

LRU 05
CARGO BAY

Quick-swap bottom rail for medical cold boxes, munition pods, or EO/IR cameras.

LRU 06
RUGGED GCS

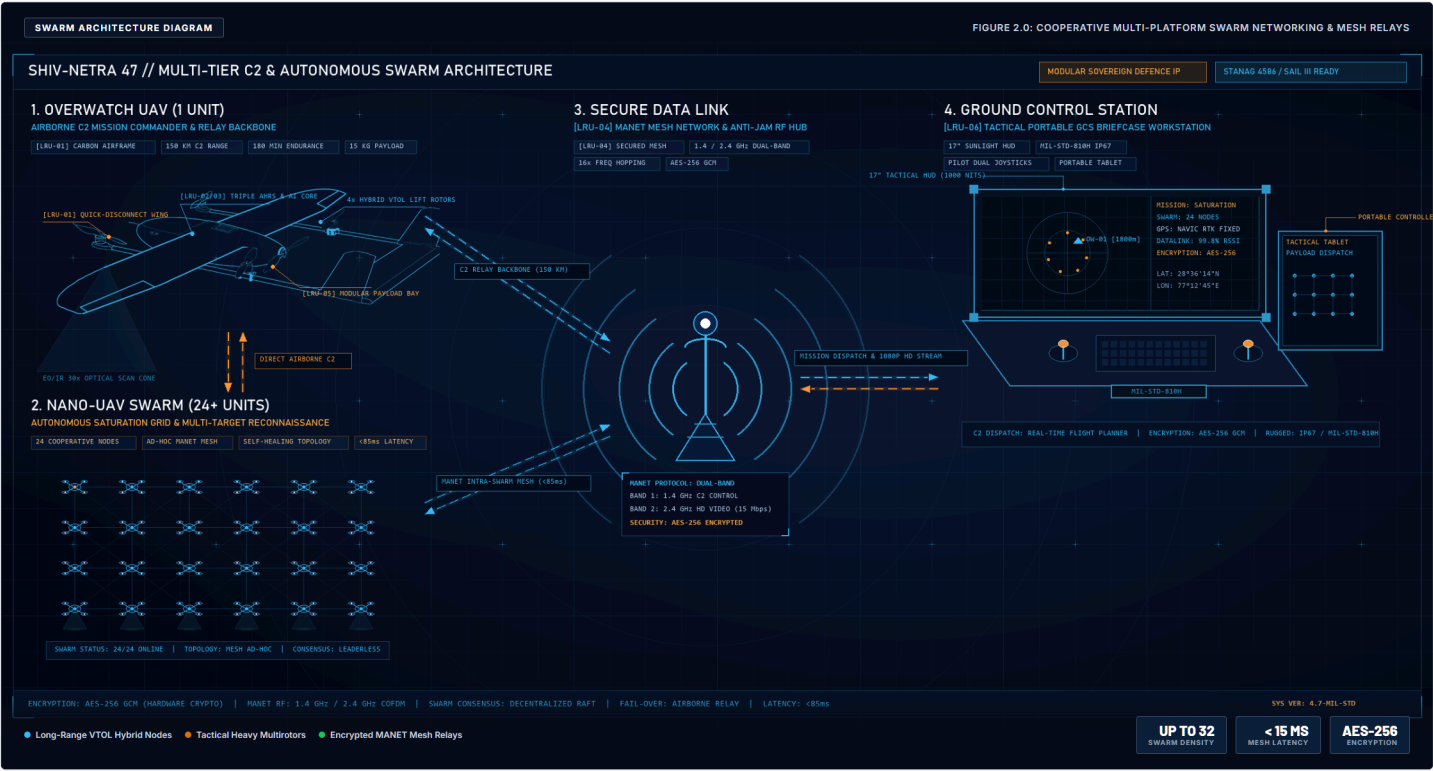
Mil-STD-883C dual-display portable command station with hands-on controls.

FORMAL ENGINEERING SPECIFICATION BENCHMARK		CERTIFICATION STANDARD: EASA SAIL III / SORA 2.5 • DOCA TYPE CERTIFICATION READY
TECHNICAL SPECIFICATION	SHIV-NETRA47 VTOL HYBRID (LONG RANGE)	SHIV-NETRA47 TACTICAL MULTIROTOR (HEAVY LIFT)
Max Takeoff Weight (MTOW)	45 kg (scalable up to 200 kg class target)	55 kg (heavy-lift tactical multirotor)
Useful Payload Capacity	5 – 15 kg (internal modular hermetic cargo pod)	15 – 25 kg (external underslung or winch payload)
Maximum Flight Endurance	Up to 180 Minutes (fixed-wing aerodynamic pusher cruise)	45 – 55 Minutes (continuous precision hovering)
Operational Command Range	150 km Line-of-Sight / Unlimited via MANET Swarm Relays	25 km Tactical Perimeter Operations
Maximum Service Ceiling	5,500 m AMSL (Himalayan passes and ridge outposts)	4,500 m AMSL (valley floor & forward trenches)
Cruise / Max Dash Speed	105 km/h cruise / 145 km/h dash speed	55 km/h transit / 0 km/h precision hover
Propulsion Architecture	Hybrid: 4x VTOL vertical lift rotors + 1x rear pusher prop	Hexacopter: 6x heavy-lift coaxial/direct-drive rotors
European Flight Heritage	Flight-proven CATUAV / BCN Drone Center European airframe	Flight-tested industrial heavy-lift multirotor platform

CAPABILITY 03 // TACTICAL SWARM INTEGRATION

MULTI-SWARM UAS ARCHITECTURE & TACTICAL INTEGRATION

Distributed Autonomous Mission Control, Self-Healing MANET Mesh Relay, and Alpine ISR



DOCTRINE 01	MESH RELAYS	DOCTRINE 02	SENSOR FUSION	DOCTRINE 03	ELECTRONIC WARFARE
SELF-HEALING MANET NETWORKING Every airborne platform acts as an active relay transceiver. If mountainous topography blocks direct line-of-sight to the ground station, the swarm dynamically re-routes command and video packets through neighboring airborne nodes without mission interruption. DYNAMIC RE-ROUTING MOUNTAIN BYPASS NO SINGLE POINT OF FAILURE		DISTRIBUTED TACTICAL ISR FUSION Multiple VTOL and multitrotor airframes observe designated forward target zones from complementary orbital angles. EO/IR video and radar tracks are correlated on-board to synthesize an integrated common operational picture (COP) for command staff. MULTI-ANGLE TRACKING AI TARGET IDENTIFICATION INTEGRATED RADAR/EO		FREQUENCY-HOPPING ECCM SUITE Hardened against adversarial jamming and spoofing across contested LAC frontiers. Employs pseudorandom frequency hopping over 1,000 hops/sec with cognitive fallback to dual NAVIC satellite constellations and optical flow dead-reckoning. 1,000+ HOPS/SEC ANTI-SPOOF NAVIC DEAD-RECKONING FAILSAFE	

CAPABILITY 01 BORDER RIDGE OVERWATCH Continuous persistent surveillance along high-altitude mountain ridgelines with seamless automated battery handoffs.	CAPABILITY 02 DEEP VALLEY RELAY Overcoming RF shadows in deep alpine ravines by positioning high-altitude VTOL hybrid nodes as overhead airborne repeaters.	CAPABILITY 03 SATURATION RESUPPLY Simultaneous synchronized multi-target delivery of emergency plasma, tactical batteries, and small arms munitions to forward squads.	CAPABILITY 04 ATTRITION SURVIVABILITY Decentralized algorithms ensure the mission continues seamlessly even if individual units are lost to terrain or electronic countermeasures.
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SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED
CIN: U30302UP2024PTC202547 • DEFENCE AEROSPACE DIVISION

Bilateral Technology Partner:
BCN Drone Center (Moia, Spain)

NATIONAL DEFENCE AEROSPACE SOVEREIGNTY

DEFENDING THE FRONTIERS OF BHARAT WITH SOVEREIGN AUTONOMOUS AIRPOWER

SOVEREIGN AIRSPACE GUARDIAN

Tactical Autonomous VTOL Resupply & Persistent High-Altitude Border Surveillance

SHIVNETRA 47

EUROPEAN SAIL III HERITAGE • MAKE IN INDIA

CORPORATE SCOPE & BUSINESS CHARTER (MOA DOMAINS)

FULL DEFENCE & AEROSPACE SPECTRUM

01 // DEFENCE SYSTEMS

AIR DEFENCE & RADAR SYSTEMS

Researching, developing, engineering, manufacturing, integrating, and commissioning radar and sensor-based surveillance, detection, tracking, C2, and countermeasure systems.

04 // R&D & ARTIFICIAL INTELLIGENCE

DEFENCE R&D & AUTONOMOUS AI

Prototyping advanced artificial intelligence, machine learning, autonomous obstacle avoidance, cyber-secure datalinks, and specialized defence electronics.

02 // UAS & DRONES

UAVS, DRONES & SWARM SYSTEMS

End-to-end design, manufacture, assembly, and integration of fixed-wing VTOLs, tactical multirotors, autonomous swarm algorithms, payloads, and flight control architectures.

05 // MRO & SUSTAINMENT

DEPOT MRO, CALIBRATION & UPGRADES

Full-lifecycle support: repair, maintenance, servicing, sensor calibration, modification, and continuous upgradation of drones, UAVs, and military aerospace assets.

03 // TRAINING & SIMULATION

DRONE PILOT & TECHNICAL TRAINING

Comprehensive pilot training, hands-on drone construction, simulation, operational procedures, and skill development programmes aligned with DGCA and European SORA standards.

06 // GLOBAL PARTNERSHIPS

TECH LICENSING & JOINT VENTURES

Procurement, distribution, technology transfer, and joint ventures in permitted aerospace products with European leaders, strictly adhering to Government of India regulations.

CORPORATE HEADQUARTERS & OPERATIONS (INDIA)

ENTITY NAME	CORPORATE IDENTIFICATION
SHIV-NETRA47 AIR DEFENCE SYSTEM PVT LTD	CIN: U30302UP2024PTC202547
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Uttar Pradesh Defence Industrial Corridor, India	IDEX Registered • GeM Portal • Startup India
OFFICIAL CONTACT	OFFICIAL PORTAL
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EUROPEAN TECHNICAL ECOSYSTEM SUPPORT

PARTNER FACILITY	REGULATORY HERITAGE
BCN Drone Center (CATUAV)	EASA SAIL III • SORA 2.5 Approved
FLIGHT TEST LOCATION	PARTNER PORTAL
Moia, Barcelona, Spain	www.bcn-drones.com
COLLABORATION SCOPE	
Bilateral UAS Design, Turnkey Assembly & High-Endurance Flight Validation	

SOVEREIGN PORTAL VERIFICATION



SCAN TO VERIFY DOSSIER
[HTTPS://SHIVNETRA47.TECH](https://shivnetra47.tech)
Official Sovereign Defence Ecosystem Portal • IDEX & Defence Procurement Documentation